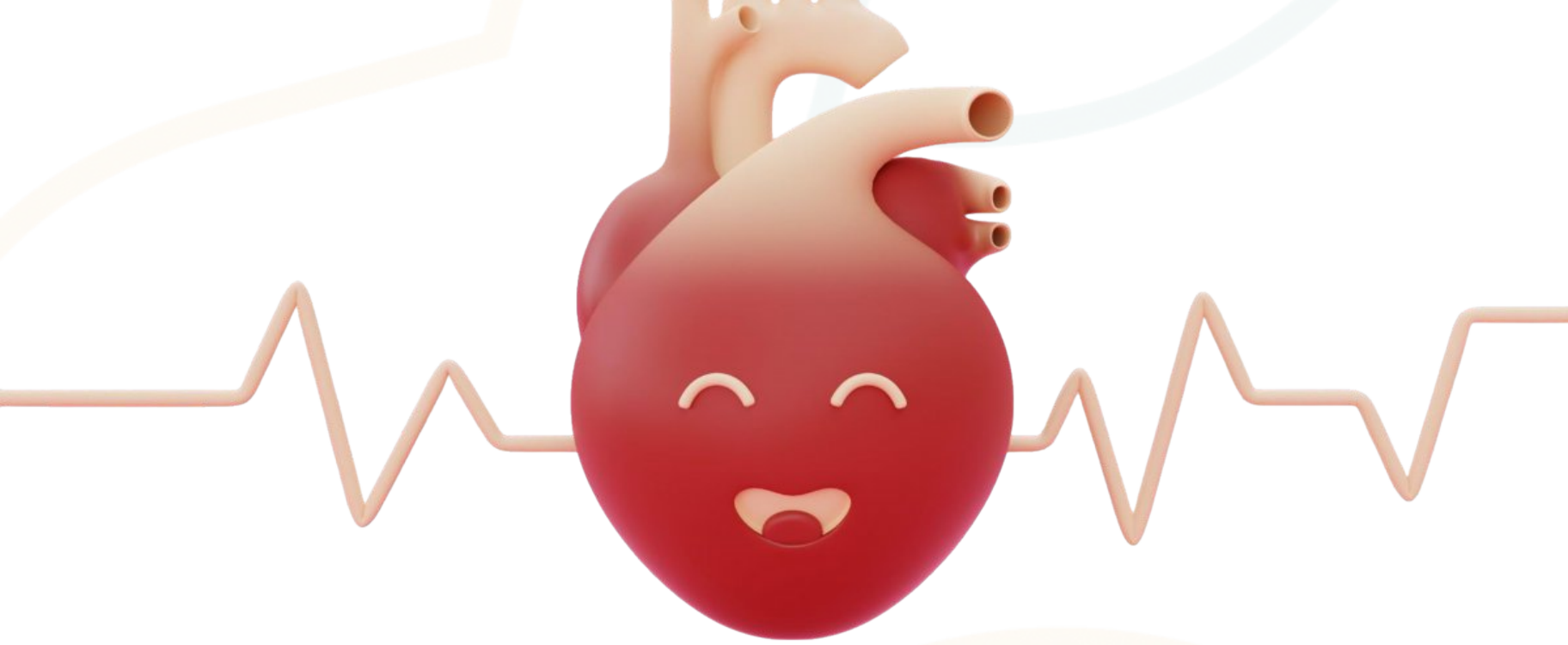
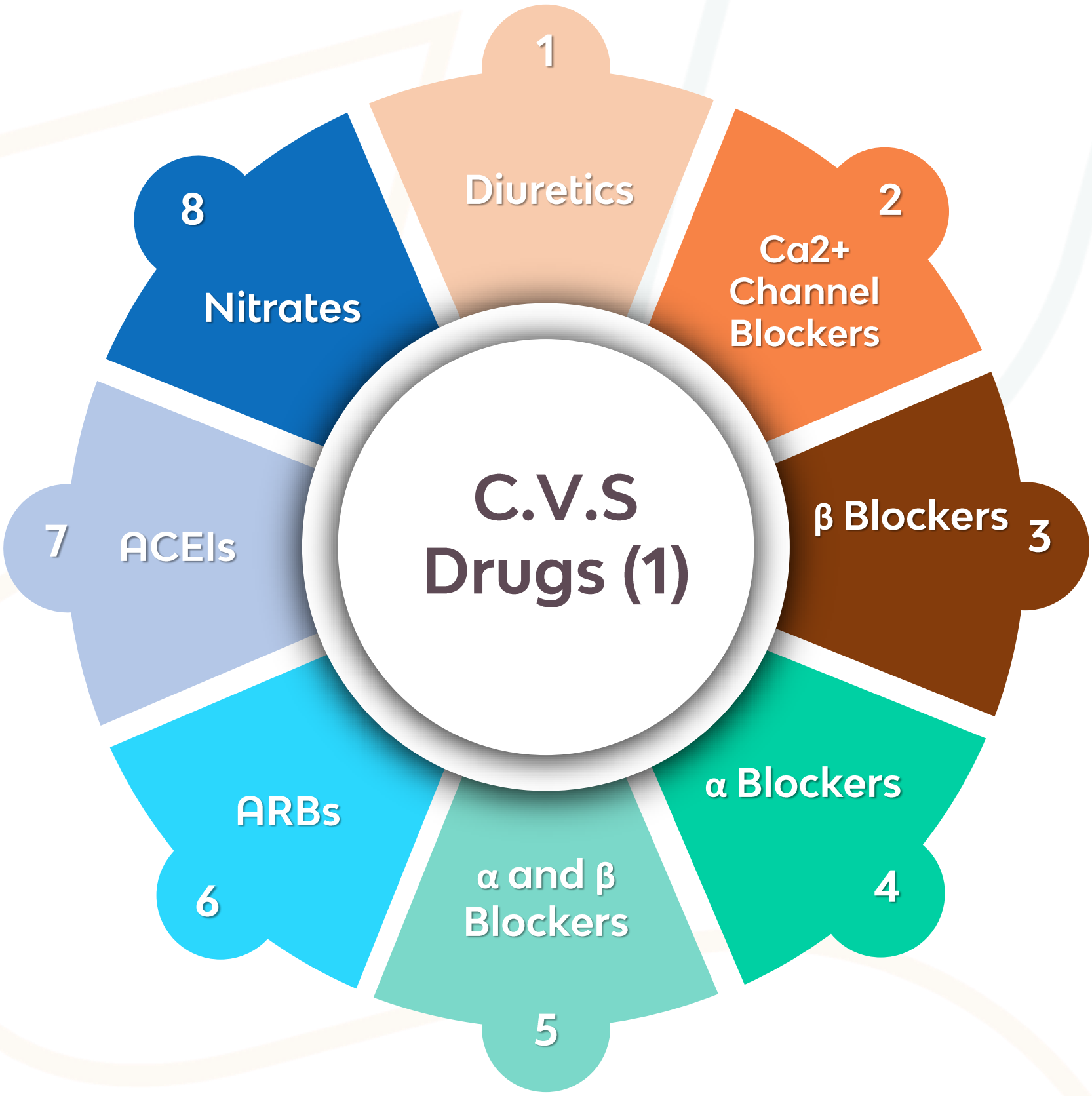
Cardiovascular Drugs (Part 1)



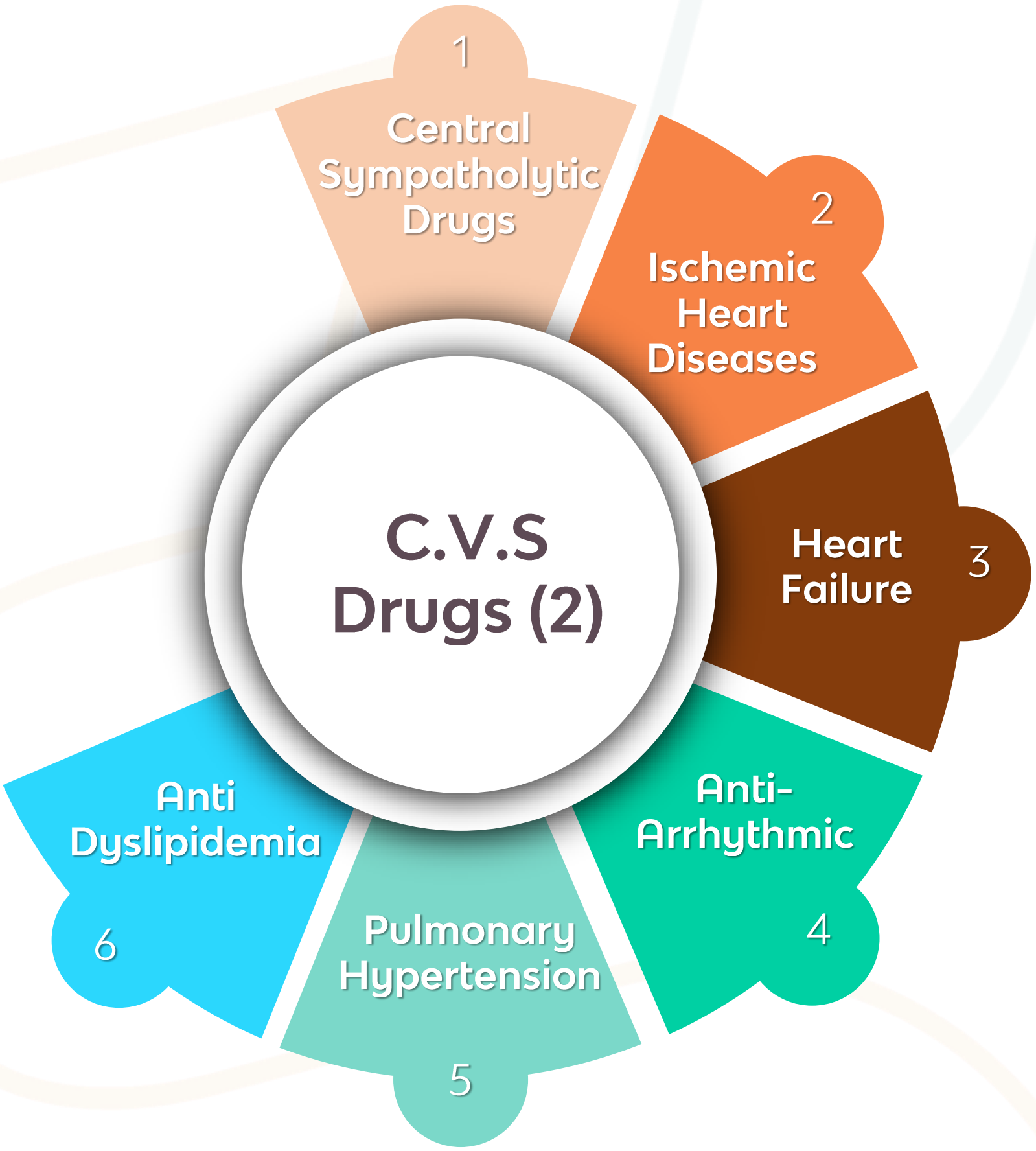
Talent Management Department

صيدليـة المتحـدة
United Pharmacy

CVS drugs classification



CVS drugs classification



Today we will learn about

01 Diuritics

02 Ca2+ Channel Blockers

03 Adrenergic Receptors Blockers

04 ACEIs

05 ARBs

06 Nitrates

صيدليـة المتحـدة
United Pharmacy

Hypertension
A chronic medical condition where blood pressure (BP) is consistently elevated (≥140/90 mmHg).

Diuretics

Increase urine output → reduce blood volume → lower blood pressure.

1. Loop Diuretics ✅
Block $Na^+/K^+/2Cl^-$ transporter in loop of Henle
Use: Edema (heart failure, kidney disease), severe hypertension

2. Thiazide Diuretics ✅
Block Na^+/Cl^- transporter in distal tubule
Use: First-line for hypertension, mild edema
Note: Better to avoid in gout.

3. Potassium-Sparing Diuretics ✅
Spironolactone: Blocks aldosterone receptors
Amiloride: Blocks Na^+ channels in distal tubule
Use: Heart failure, resistant hypertension, hypokalemia prevention
Note: Spironolactone may cause gynecomastia

1

Diuretics

K+ Sparing Diuretics

- ALDACTONE® | *Spironolactone*

Loop Diuretics

- LASIX® | *Furosemide*

Thiazide Diuretics

- MONOZIDE ® | *Hydrochlorothiazide*
- NATRILIX® | *Indapamide*

Calcium Channel Blockers (CCBs)

Block calcium entry into heart and blood vessel muscle cells → relax vessels, slow heart rate (depending on type)

1. Vaso-selective (Dihydropyridines) ✅
Use: Hypertension, angina (chronic stable)
Note: May cause ankle swelling, flushing, constipation.

2. Cardio-selective (Non-Dihydropyridines) ✅
Use: Hypertension, arrhythmias, angina
Note: Can cause bradycardia; avoid with beta-blockers.

3. Non-Selective ✅

2 Ca2+ Channel Blockers

Vaso-Selective

- *VALERCADIP ®| Lercanidipine*
- *SCODIPINE ® & LODIPAM ®| Amlodipine*
- *ADALAT LA ®| EPILAT®| Nifedipine*

Non-Selective

- *STUGERON®| Cinnarizine*

Cardio-Selective

- *ISOPTIN ®| Verapamil*
- *Dilzem Ret ®| Diltiazem*

Adrenergic Receptors Blockers

1. **Beta Blockers** ✅
   a. Selective (β1-blockers)
   They block β1 receptors (mainly in the heart)
   Use: Hypertension, angina, heart failure, arrhythmias
   Note: Safer in asthma than non-selective; avoid sudden stop
   b. Non-Selective (β1 & β2 blockers)
   They block both β1 (heart) & β2 (lungs, vessels)
   Use: Migraine, tremors, portal hypertension
   Note: Avoid in asthma/COPD (can cause bronchospasm)

2. **Alpha Blockers (α1-blockers)** ✅
   They block α1 receptors → relax blood vessels
   Use: Hypertension, Benign Prostatic Hyperplasia (BPH)

3. **Mixed Alpha & Beta Blockers** ✅
   They block β1, β2, and α1 receptors → ↓ HR & vascular resistance
   Note: Labetalol: Safe in pregnancy-related hypertension
   Carvedilol: Used in heart failure

3 **β Blockers**

- **Selective**
  - NORMOTEN ® | *Atenolol*
  - CONCOR ® | *Bisoprolol*
  - SELOKENZOC ® | *Metoprolol*
- **Selective & Vasodilator**
  - NEBILET ® | *Nebivolol*
- **Non-Selective**
  - INDERAL® | *Propranolol*

4 **α Blockers**

- CARDURA® | *Doxazosin*

5 **α & β Blockers**

- RAVILDO®| *Carvedilol*

Drugs affecting RAAS

(Renin-Angiotensin-Aldosterone System)
A hormone system that regulates blood pressure and fluid balance. Overactivation can lead to hypertension, which is managed by drugs like ACEIs and ARBs.

1. ACE Inhibitors (ACEIs) 🧪

Block the Angiotensin-Converting Enzyme → ↓ angiotensin II → vasodilation + ↓ water retention

Note:

- May cause dry cough.
- Contraindicated in pregnancy

2. Angiotensin II Receptor Blockers (ARBs) 🧪

Block angiotensin II receptors → similar effects as ACEIs without increasing bradykinin.

Note:

- No cough
- Avoid in pregnancy

6

ACEIs

- CAPOCARD ® | *Captopril*
- ZESTRIL ® | *Lisinopril*
- RAMIPRIL ® | *Ramipril*

7

ARBs

- MICARDS ® | *Telmisartan*
- SARTAN ® | *Losartan*
- TABUVAN ® | *Valsartan*
- APROVEL ® | *Irbesartan*

Nitrates

Dilate blood vessels (mainly veins) → ↓ preload → ↓ heart workload.

Short-Acting Nitrates ✅

Use: Acute angina attacks (chest pain relief within minutes)

Note:

- Works quickly (2–5 minutes)
- Sit down before using (may cause dizziness)
- Store in a cool, dark place; short shelf life

Long-Acting Nitrates ✅

Use: Prevention of angina
Adjunct in chronic heart failure

Caution: Do NOT combine with PDE5 inhibitors (e.g., sildenafil) → risk of dangerous hypotension

8

Nitrates

Long Acting

- Nitroderm ®| *Glyceryl trinitrate*

Short Acting

- Isobide ®| *Isosorbide dinitrate*

الآن حان دور الأنشطة
اضغط على الزر للبدء



ابدأ النشاط